# Rigorous Proof of Equivalence of Two Limits

Johar M. Ashfaque

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## Introduction

In this note, we evaluate the two limits provided in the images and rigorously prove their equivalence. The two limits are defined as follows:

$$L_1 = \lim_{x \to \infty} \frac{x - e^{\sqrt{x}}}{x + e^{\sqrt{x}}} \tag{1}$$

$$L_2 = \lim_{x \to 0} \frac{e^{\tan(x)} - e^{\sin(x)}}{e^{\tan^{-1}(x)} - e^{\sin^{-1}(x)}} \tag{2}$$

## Evaluation of $L_1$

Let us consider the first limit:

$$L_1 = \lim_{x \to \infty} \frac{x - e^{\sqrt{x}}}{x + e^{\sqrt{x}}}$$

To resolve the indeterminate form $\frac{\infty}{\infty}$, we factor out the dominant term, $e^{\sqrt{x}}$, from both the numerator and the denominator:

$$L_1 = \lim_{x \to \infty} \frac{e^{\sqrt{x}} \left( \frac{x}{e^{\sqrt{x}}} - 1 \right)}{e^{\sqrt{x}} \left( \frac{x}{e^{\sqrt{x}}} + 1 \right)} = \lim_{x \to \infty} \frac{\frac{x}{e^{\sqrt{x}}} - 1}{\frac{x}{e^{\sqrt{x}}} + 1}$$

We now evaluate the limit of the quotient $\frac{x}{e^{\sqrt{x}}}$. Let $u = \sqrt{x}$. As $x \to \infty$, $u \to \infty$. The expression becomes:

$$\lim_{x \to \infty} \frac{x}{e^{\sqrt{x}}} = \lim_{u \to \infty} \frac{u^2}{e^u}$$

Since this is an $\frac{\infty}{\infty}$ indeterminate form, we can apply L'Hôpital's rule twice:

$$\lim_{u \to \infty} \frac{u^2}{e^u} \overset{\text{L'H}}{=} \lim_{u \to \infty} \frac{2u}{e^u} \overset{\text{L'H}}{=} \lim_{u \to \infty} \frac{2}{e^u} = 0$$

Substituting this result back into the expression for $L_1$, we get:

$$L_1 = \frac{0 - 1}{0 + 1} = -1$$

## Evaluation of $L_2$

Now we consider the second limit:

$$L_2 = \lim_{x \to 0} \frac{e^{\tan(x)} - e^{\sin(x)}}{e^{\tan^{-1}(x)} - e^{\sin^{-1}(x)}}$$

We begin by factoring $e^{\sin(x)}$ from the numerator and $e^{\sin^{-1}(x)}$ from the denominator:

$$L_2 = \lim_{x \to 0} \frac{e^{\sin(x)} \left(e^{\tan(x)-\sin(x)} - 1\right)}{e^{\sin^{-1}(x)} \left(e^{\tan^{-1}(x)-\sin^{-1}(x)} - 1\right)}$$

As $x \to 0$, we have $\sin(x) \to 0$ and $\sin^{-1}(x) \to 0$, which implies $e^{\sin(x)} \to 1$ and $e^{\sin^{-1}(x)} \to 1$. Thus, the limit simplifies to:

$$L_2 = \lim_{x \to 0} \frac{e^{\tan(x)-\sin(x)} - 1}{e^{\tan^{-1}(x)-\sin^{-1}(x)} - 1}$$

Using the standard limit $\lim_{u \to 0} \frac{e^u - 1}{u} = 1$, we know that $e^u - 1 \sim u$ as $u \to 0$. Since both $(\tan(x) - \sin(x)) \to 0$ and $(\tan^{-1}(x) - \sin^{-1}(x)) \to 0$ as $x \to 0$, we can replace the exponential terms with their asymptotic equivalents:

$$L_2 = \lim_{x \to 0} \frac{\tan(x) - \sin(x)}{\tan^{-1}(x) - \sin^{-1}(x)}$$

To evaluate this new limit, we employ the Maclaurin series expansions around $x = 0$ for the trigonometric functions up to the third degree:

$$\begin{aligned}
\tan(x) &= x + \frac{x^3}{3} + \mathcal{O}(x^5) \\
\sin(x) &= x - \frac{x^3}{6} + \mathcal{O}(x^5) \\
\tan^{-1}(x) &= x - \frac{x^3}{3} + \mathcal{O}(x^5) \\
\sin^{-1}(x) &= x + \frac{x^3}{6} + \mathcal{O}(x^5)
\end{aligned}$$

Using these expansions, we calculate the leading-order behavior of the numerator and denominator:

$$\begin{aligned}
\tan(x) - \sin(x) &= \left(x + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) + \mathcal{O}(x^5) = \left(\frac{1}{3} + \frac{1}{6}\right) x^3 + \mathcal{O}(x^5) = \frac{x^3}{2} + \mathcal{O}(x^5) \\
\tan^{-1}(x) - \sin^{-1}(x) &= \left(x - \frac{x^3}{3}\right) - \left(x + \frac{x^3}{6}\right) + \mathcal{O}(x^5) = \left(-\frac{1}{3} - \frac{1}{6}\right) x^3 + \mathcal{O}(x^5) = -\frac{x^3}{2} + \mathcal{O}(x^5)
\end{aligned}$$

Substituting these approximations back into the limit expression:

$$L_2 = \lim_{x \to 0} \frac{\frac{x^3}{2} + \mathcal{O}(x^5)}{-\frac{x^3}{2} + \mathcal{O}(x^5)} = \lim_{x \to 0} \frac{\frac{1}{2} + \mathcal{O}(x^2)}{-\frac{1}{2} + \mathcal{O}(x^2)} = \frac{1/2}{-1/2} = -1$$

# Conclusion

Our evaluations yielded $L_1 = -1$ and $L_2 = -1$. Therefore, we have established the equivalence of the two limits:

$$\lim_{x \to \infty} \frac{x - e^{\sqrt{x}}}{x + e^{\sqrt{x}}} = \lim_{x \to 0} \frac{e^{\tan(x)} - e^{\sin(x)}}{e^{\tan^{-1}(x)} - e^{\sin^{-1}(x)}}$$